

Document Processing and AI Query Generation Script

1. Introduction

This script processes text-based documents (PDF, DOCX, TXT) and interacts with an AI model to generate responses based on the extracted content. It takes a document file as input, extracts its content, and then queries an AI model to generate a response.

2. Why Use This Script?

- Automates text extraction from different file formats.
- Provides AI-generated insights based on the document content.
- Saves time in manually reviewing lengthy documents.
- Supports multiple file formats: PDF, DOCX, and TXT.

3. How It Works

The script follows these steps:

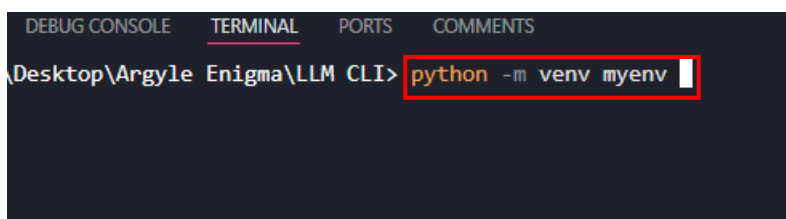
1. **Read the input file:** Extracts text from the given PDF, DOCX, or TXT file.
2. **Process the extracted text:** Formats it for AI processing.
3. **Interact with an AI model:** Uses an AI model to generate a response based on the extracted content and user query.
4. **Display the response:** Outputs the AI-generated text to the console.

4. Step-by-Step Approach

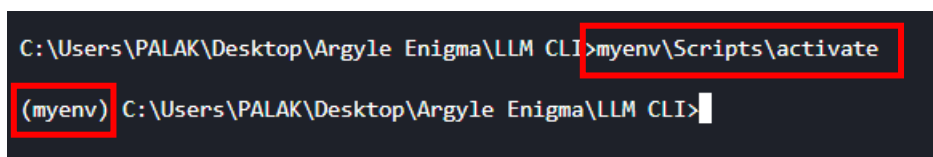
Step 1: Create a Virtual Environment

It is recommended to run the script inside a virtual environment to manage dependencies effectively. Use the following commands to create and activate a virtual environment:

1) Command for Creation of Virtual Environment

A terminal window with a dark background. At the top, there are four tabs: 'DEBUG CONSOLE', 'TERMINAL' (which is active), 'PORTS', and 'COMMENTS'. The terminal shows the command 'python -m venv myenv' being entered at the prompt 'Desktop\Argyle Enigma\LLM CLI>'. The command and the prompt are highlighted with a red rectangular box.

2) Command for activate the virtual environment

A terminal window with a dark background. The terminal shows the command 'myenv\Scripts\activate' being entered at the prompt 'C:\Users\PALAK\Desktop\Argyle Enigma\LLM CLI>'. The command is highlighted with a red rectangular box. On the next line, the prompt has changed to '(myenv) C:\Users\PALAK\Desktop\Argyle Enigma\LLM CLI>', and this new prompt is also highlighted with a red rectangular box.

Step 2: Install Dependencies

Ensure you have the required Python libraries installed to handle document processing and AI interaction. Run the following command:

You can find the requirements.txt in below link. (make during execution give proper path for requirement.txt according to your folder or else you can use relative path also)

My current directory is C:\Users\PALAK\Desktop\Argyle Enigma\LLM CLI. But Adjust the folder structure as you want.

Link for Resources : [Click Here](#)

```
(myenv) C:\Users\PALAK\Desktop\Argyle Enigma\LLM CLI>pip install -r requirements.txt
Collecting ollama (from -r requirements.txt (line 1))
  Obtaining dependency information for ollama from https://files.pythonhosted.org/packages/31/83/c3ffac86906c10184c88c2e916460806b072a2cfe34cdcaf3a0c0e836d39/ollama-0.4.7-py3-none-any.whl.metadata
  Using cached ollama-0.4.7-py3-none-any.whl.metadata (4.7 kB)
Collecting PyPDF2 (from -r requirements.txt (line 2))
  Obtaining dependency information for PyPDF2 from https://files.pythonhosted.org/packages/8e/5e/c86a5643653825d3c913719e788e41386bee415c2b87b4f955432f2de6b2/pypdf2-3.0.1-py3-none-any.whl.metadata
  Using cached pypdf2-3.0.1-py3-none-any.whl.metadata (6.8 kB)
Collecting python-docx (from -r requirements.txt (line 3))
  Obtaining dependency information for python-docx from https://files.pythonhosted.org/packages/3e/3d/330d9efbdb816d3f60bf2ad92f05e1708e4a1b9abe80461ac3444c83f749/python_docx-1.1.2-py3-none-any.whl.metadata
```

Step 3: Execution of the script.

To execute the script, run the following command format in the terminal:

```
(myenv) C:\Users\PALAK\Desktop\Argyle Enigma\LLM CLI>python <python_file_name> <document_name> <query>
```

For example:

You can find the main.py and Sample.pdf files in the link below. And make sure you give correct path for main.py and sample.pdf according to your folder structure.

My current directory is C:\Users\PALAK\Desktop\Argyle Enigma\LLM CLI. But Adjust the folder structure as you want.

Link for Resources : [Click Here](#)

```
(myenv) C:\Users\PALAK\Desktop\Argyle Enigma\LLM CLI>python main.py sample.pdf "Summerize this document"
```

This command extracts text from sample.pdf, processes it, and sends the query to the AI model for generating a response.